

Brain Tumors

Cause

We do not know exactly what causes brain tumors. Some breeds are at higher risk, so genetics and environment probably both play a role.

Signs

Signs depend on which part of the brain is affected:

- **Seizures** (in dogs, sometimes the only sign)
- Confusion, sleepiness, behavior change
- Poor balance, weakness
- Pain, if the tumor irritates the brain's covering (meninges)

As the tumor grows, signs worsen and can progress to unresponsiveness, coma and death.

Diagnosis

- **Brain MRI** to find and characterize the mass
- Blood tests and **spinal fluid analysis** to rule out infection or inflammation

Treatment

To shrink or remove the tumor: surgery, radiation therapy, chemotherapy. **To control signs:** steroids, pain medication and anti-seizure medication (often needed when the front of the brain is involved).

How do steroids help?

Steroids reduce **brain swelling** and pressure, which can improve signs quickly. We often start with injectable dexamethasone, then switch to oral prednisone/prednisolone, reduced to the lowest effective dose. **Steroids do not shrink most tumors** (the

exception is CNS lymphoma). With steroids alone, survival is typically 1–3 months. Steroids are also used before and after surgery.

What to expect — meningioma

Meningioma is the most common brain tumor in dogs and cats. Most pets keep a good quality of life until the late stages. Approximate survival in **dogs**:

Treatment	Typical survival
Steroids + seizure medication	~3 months
Adding oral chemotherapy (hydroxyurea)	~6 months
Surgery + hydroxyurea, or radiation alone	~1 year
Surgery + radiation	~1½ years

In **cats**, surgery is usually recommended: about **95%** return to normal and recurrence is low (~15%). In dogs, surgery is less often curative and radiation is often recommended after or instead of surgery.

Other tumor types

More aggressive cancers (histiocytic sarcoma, nasal carcinoma, lymphoma) carry a poor to grave prognosis — up to about 8 months at best, even with radiation. Clinical trials may be an option; ask for our *Brain Tumor Clinical Trials* handout.

When to call us

Seizures that are not controlled, sudden worsening, not eating, or signs of pain. Many families choose euthanasia when seizures can no longer be controlled or signs suddenly worsen — we are here to help you with that decision.